Improve One Cause

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

WORKBOOK · 1 OF 21

Recall and choose your starting point

> Record your starting explanation for Improve One Cause before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Test Edge Cases and Fairness

Without opening the old lesson, explain one key idea from Test Edge Cases and Fairness. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Build a Paper Prototype

Without opening the old lesson, explain one key idea from Build a Paper Prototype. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Find a Real Problem

Without opening the old lesson, explain one key idea from Find a Real Problem. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

WORKBOOK · 2 OF 21

Practise with fading help

> Complete the supported Improve One Cause trace and record the first step you can explain without help.

Use W2: Check the **repair** and earlier behaviour for Improving and checking **regression**s. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

WORKBOOK · 3 OF 21

Individual reasoning check

> Check the proposed Improve One Cause explanation against the supplied facts before discussing it.

Use the worked case for Improving and checking **regression**s. Proposed explanation: “One **repair**ed example proves improvement everywhere”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your **repair** if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 21

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Audit a different proposed repair

Equipment catalogue E lists E1 first with capacity 4 and E2 second with capacity 8. Valid whole group sizes are 1 to 10. Choose the first capacity at least the group size, or no match; invalid **input** needs clarification.

Old U0 uses strictly greater capacity and stops at the first fit. Proposed U1 includes equality but keeps searching and returns the last fit. Independently compare U0, U1 and the requirement for groups 8 and 3.

Identify the **repair** and **regression**, propose U2 and give one validation check. No booking occurs.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 21

Your learning target

> Connect the Improve One Cause target to a trace and a justified conclusion.

Version A makes four errors. Version B makes one after lighting, labels and rules all change. What can you conclude, and how would you isolate the effect of lighting?

WORKBOOK · 6 OF 21

Solve the given case

> Solve the supplied Improve One Cause case and show the deciding evidence.

The team changes lighting, labels and rules together; errors fall from four to one.

> Can it identify the cause of improvement?

Show your trace, calculation or **evidence**

> **Write or draw your reasoning and evidence.**

Check a changed case

> Apply the procedure to the changed case and record why the conclusion follows.

Dim photos fail. You change the lighting and all decision rules together; performance improves. Can you tell which change helped? Design a clearer comparison.

My independent answer and reason

Write or draw your reasoning and evidence.

Week 35 Test and reflect

WORKBOOK · 8 OF 21

Design a revealing test

> Specify a changed input, expected result and observation separately.

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

WORKBOOK · 9 OF 21

Project checkpoint

> Justify what Improve One Cause adds to Helpful Classroom Sorter and keep the evidence.

Create sorter version 2 with one **evidence**-linked change and a before/after test table.

The project change and **evidence** I will keep

> **Write or draw your reasoning and evidence.**

WORKBOOK · 10 OF 21

Self check

> Judge explanation, evidence, transfer and limits separately; attach the work behind your judgement.

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

WORKBOOK · 11 OF 21

Transfer practice from the original programme

> Use this earlier case as transfer practice. A case already discussed with feedback is not fresh assessment evidence.

> Version A makes four errors. Version B makes one after lighting, labels and rules all change. What can you conclude, and how would you isolate the effect of lighting?

> **My independently reasoned answer**

Record support used: none, hint or teacher explanation.

W1 Write the change record

> Write the change record. Apply the supplied Improve One Cause facts and keep your reasoning.

Use S0's failure on 10 Yes with catalogue B version 1. Record the expected and incorrect **output**s, the faulty time comparison, the exact S1 correction and the other conditions that must remain.

Explain what old **evidence** to preserve.

My reasoning and evidence

WORKBOOK · 13 OF 21

W2 Check the repair and earlier behaviour

> Check the repair and earlier behaviour. Apply the supplied Improve One Cause facts and keep your reasoning.

Plan and operate the S1 **repair** and **regression** set described in the lesson, using the appropriate catalogue version for each case. Record expected and observed results separately with reasons and traces.

Summarise full passes over all attempted cases, list unrun cases separately and identify any remaining failures. Do not replace observed mistakes with worked answers.

My reasoning and evidence

WORKBOOK · 14 OF 21

W3 Reject a repair that breaks quiet requests

> Reject a repair that breaks quiet requests. Apply the supplied Improve One Cause facts and

keep your reasoning.

T1 accepts equal times but omits the quiet check. Explain why it passes 10 Yes yet fails 9 Yes. Use S0's earlier correct result for 9 Yes to justify the word **regression**.

What should change in T1, and which two cases should be rerun first?

My reasoning and evidence

WORKBOOK · 15 OF 21

W4 Audit a different proposed repair

Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Audit a different proposed repair. Apply the supplied Improve One Cause facts and keep your reasoning.

Equipment catalogue E lists E1 first with capacity 4 and E2 second with capacity 8. Valid whole group sizes are 1 to 10. Choose the first capacity at least the group size, or no match; invalid **input** needs clarification.

Old U0 uses strictly greater capacity and stops at the first fit. Proposed U1 includes equality but keeps searching and returns the last fit. Independently compare U0, U1 and the requirement for groups 8 and 3.

Identify the **repair** and **regression**, propose U2 and give one validation check. No booking occurs.

My reasoning and evidence

WORKBOOK · 16 OF 21

W5 Protect the meaning of final evaluation

> Protect the meaning of final evaluation. Apply the supplied Improve One Cause facts and keep your reasoning.

After seeing a reserved final case fail, a team edits its candidate, reruns that same case and calls the result “a successful unseen test of the new version”. Explain what is wrong and what should be recorded instead.

State what must be saved before an evaluation begins.

My reasoning and evidence

WORKBOOK · 17 OF 21

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

WORKBOOK · 18 OF 21

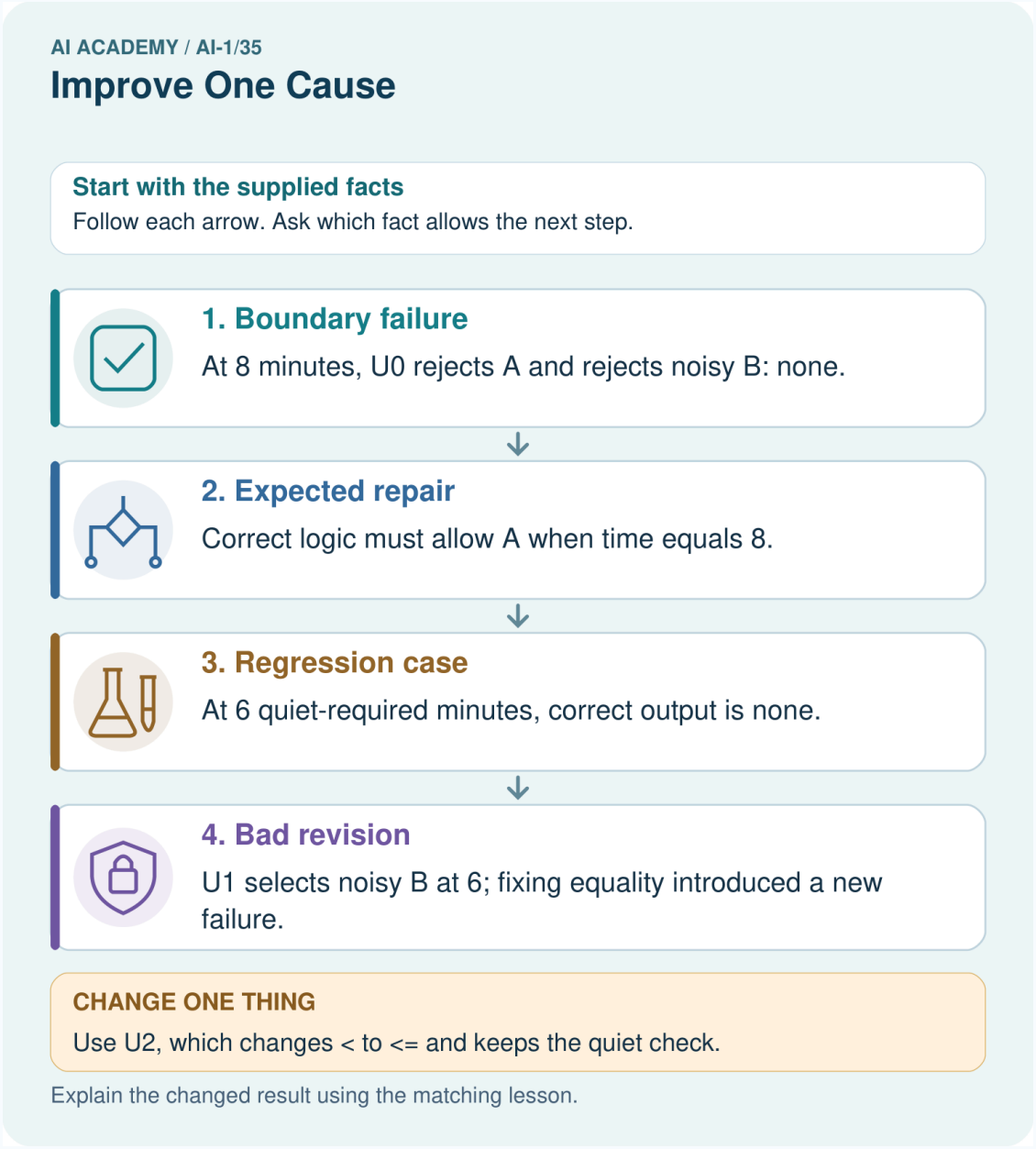
Pictorial case practice: Improve One Cause

> A fictional quiet-activity chooser has a boundary bug and a proposed repair. Explain the labelled step before checking the changed condition.

A fictional quiet-activity chooser has a boundary bug and a proposed **repair**.

A: quiet Yes, 8 minutes; B: quiet No, 5 minutes. Request quiet required Yes.

U0 accepts only duration<available time. U1 fixes ≤ but accidentally removes the quiet requirement.



Follow the labelled picture with the matching explanation. Open the vector diagram

Explain why “our **repair**ed version now picks something” is not enough to show success at 6.

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: Maintain sorter **regression** tests and compare complete version traces, retaining new failures rather than hiding them.

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Helpful Classroom Sorter record.

Project: Helpful Classroom Sorter

Current milestone: Present and defend. Demonstrate the system, explain one failure and justify the next improvement with **evidence**.

Maintain sorter **regression** tests and compare complete version traces, retaining new failures rather than hiding them.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

WORKBOOK · 20 OF 21

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional quiet-activity chooser has a boundary bug and a proposed **repair**.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? U1 selects noisy B at 6; fixing equality introduced a new failure.

1. Bad revision

2. **Regression** case

3. Expected **repair**

4. Boundary failure

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Bad revision U1 selects noisy B at 6; fixing equality introduced a new failure. A **repair** must preserve the full contract, not just make one failing example pass.

The quiet condition and duration comparison are separate requirements. Changing only the inequality targets the recorded cause; removing the quiet check changes another behavior.

Save previously passing cases to reveal such **regression**s.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Use U2, which changes < to ≤ and keeps the quiet check. Which conclusion is supported by the supplied facts?

1. At 8 return A; at 6 return none.

2. This one example guarantees correct results for every future **input**.

3. We can decide the result without examining the changed **input** or the procedure.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: At 8 return A; at 6 return none.

> Equality is **repair**ed while B remains ineligible because it is noisy.

Repair a missing building block

Review Test Edge Cases and Fairness (ai-1/34). Goal: Choose tests just below, at and above a decision boundary. Short practice: Which single boundary run most directly detects the strict-comparison bug, and why?.

Return to this week and explain what you changed.

Review Build a Paper Prototype (ai-1/32). Goal: Build a runnable paper prototype with explicit instructions and visible state. Short practice: What should appear in the test log before and after the **repair**?.

Return to this week and explain what you changed.

Review Find a Real Problem (ai-1/29). Goal: Describe a specific user difficulty using supplied **evidence** without inventing its cause.

Short practice: Write a one-sentence project goal and say what **evidence** would count as a failure.. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

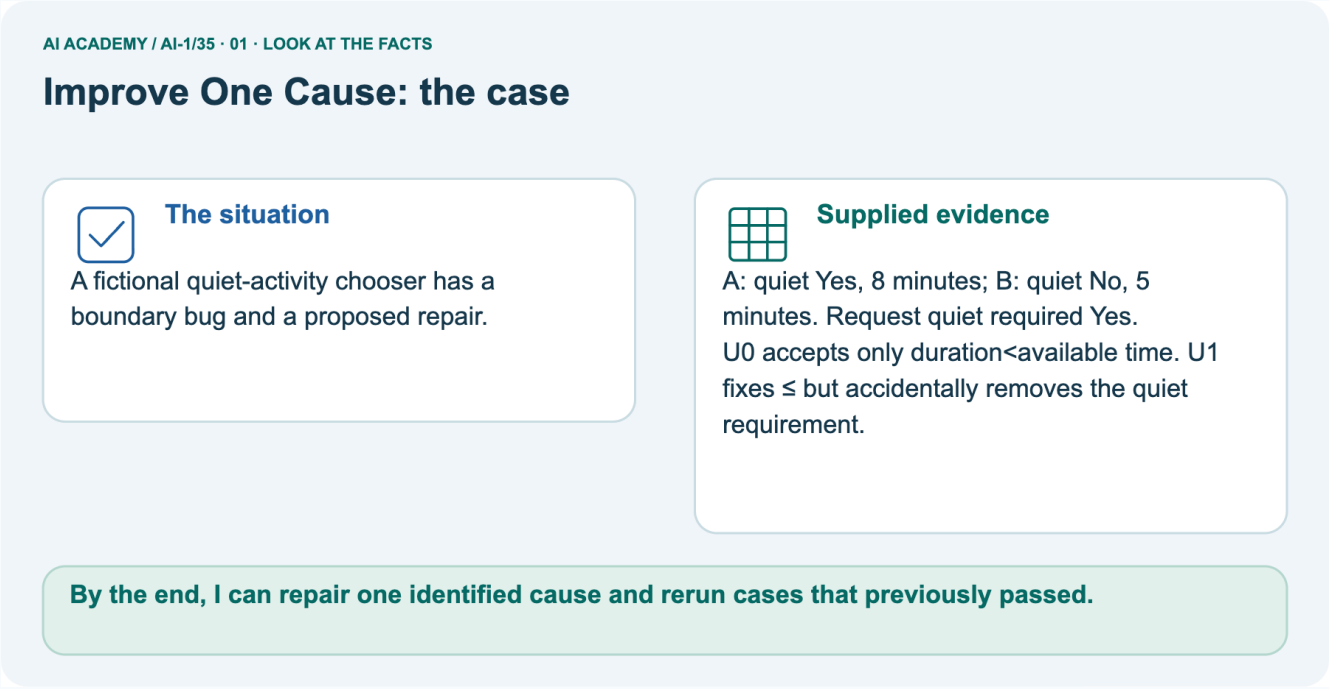
Drafts stay in this browser when storage is available; clear drafts on a shared device.

> **What I repaired, the evidence I kept and the next test**

Visual concept journey

> Explain each Improve One Cause picture from its labels, then record a relevant limit.

01. Improve One Cause: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

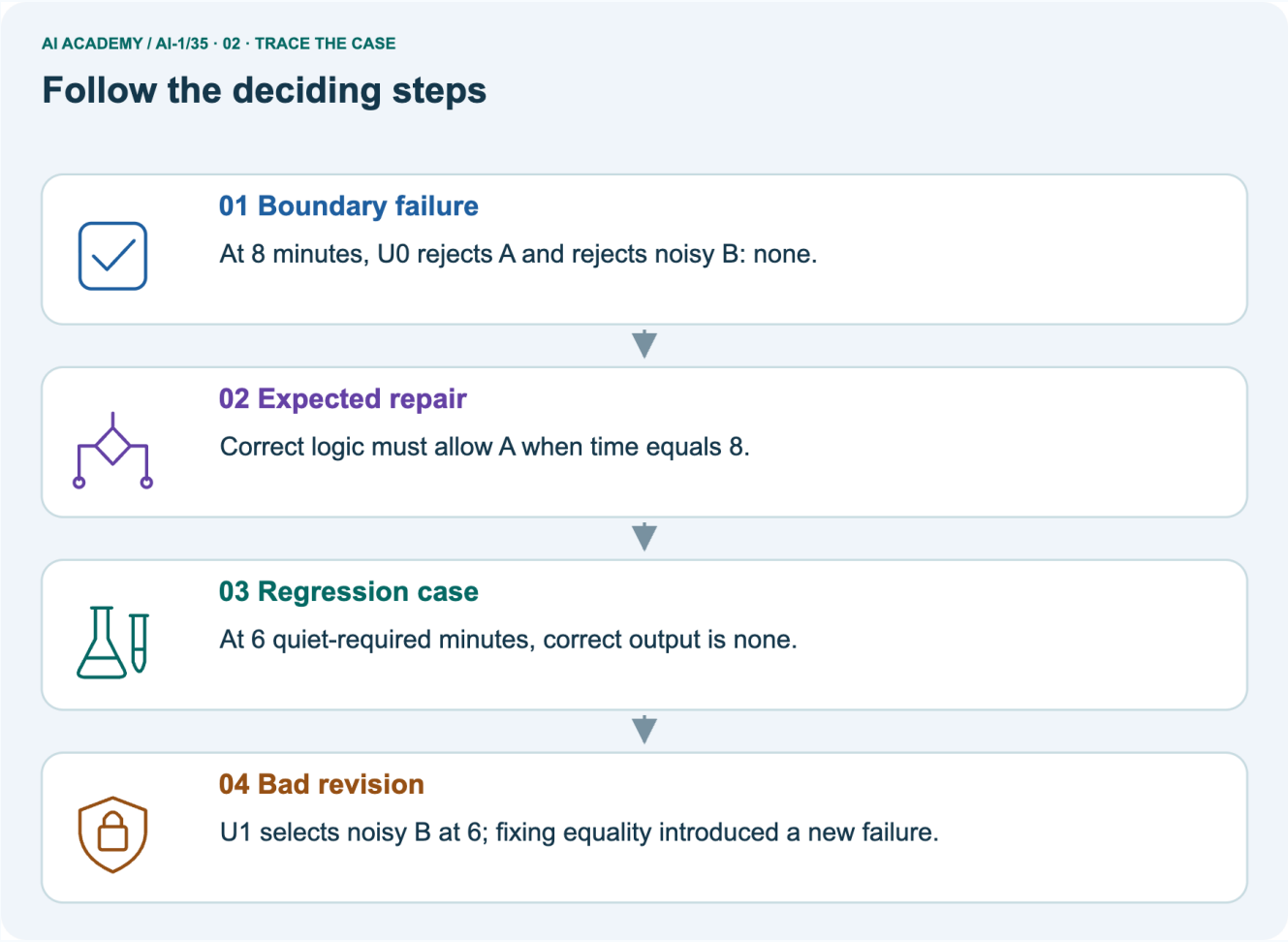
A fictional quiet-activity chooser has a boundary bug and a proposed **repair**. Supplied facts: A: quiet Yes, 8 minutes; B: quiet No, 5 minutes. Request quiet required Yes. U0 accepts only duration<available time.

U1 fixes ≤ but accidentally removes the quiet requirement.

> Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



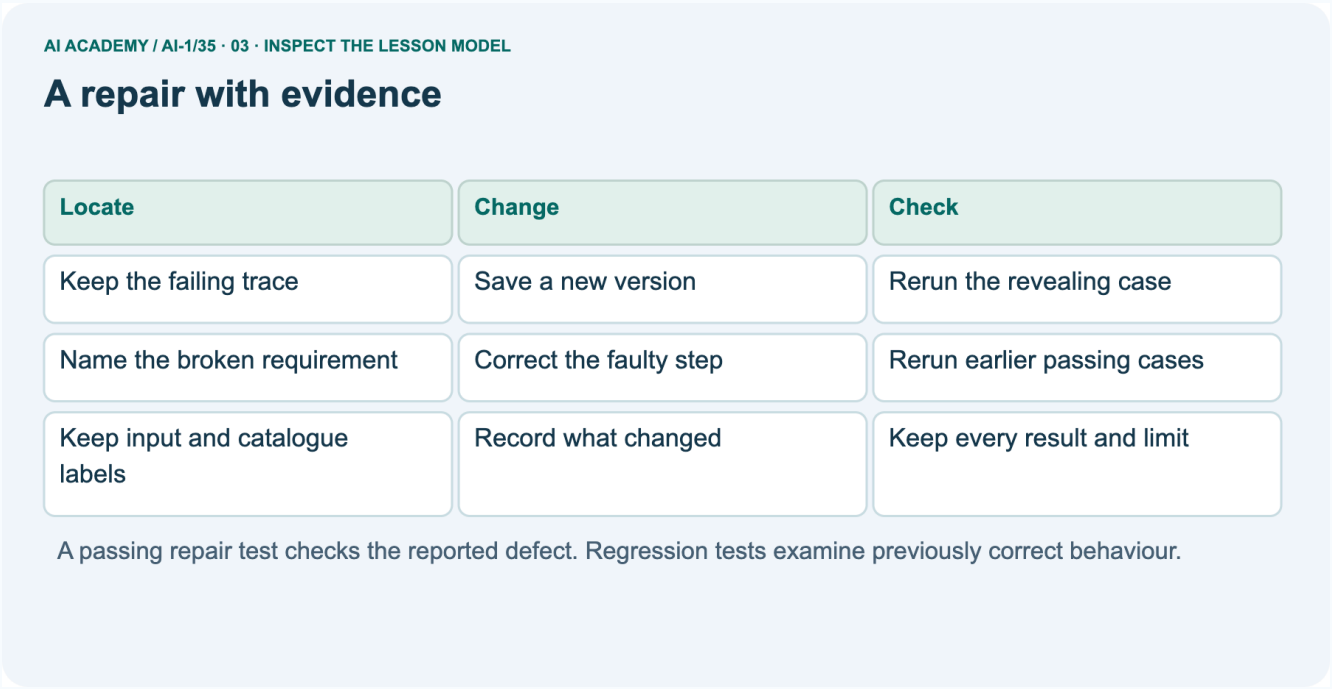
Follow the labelled picture with the matching explanation. Open the vector diagram

Boundary failure: At 8 minutes, U0 rejects A and rejects noisy B: none. → Expected **repair**: Correct logic must allow A when time equals 8. → **Regression** case: At 6 quiet-required minutes, correct **output** is none. → Bad revision: U1 selects noisy B at 6; fixing equality introduced a new failure.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. A repair with evidence



Follow the labelled picture with the matching explanation. Open the vector diagram

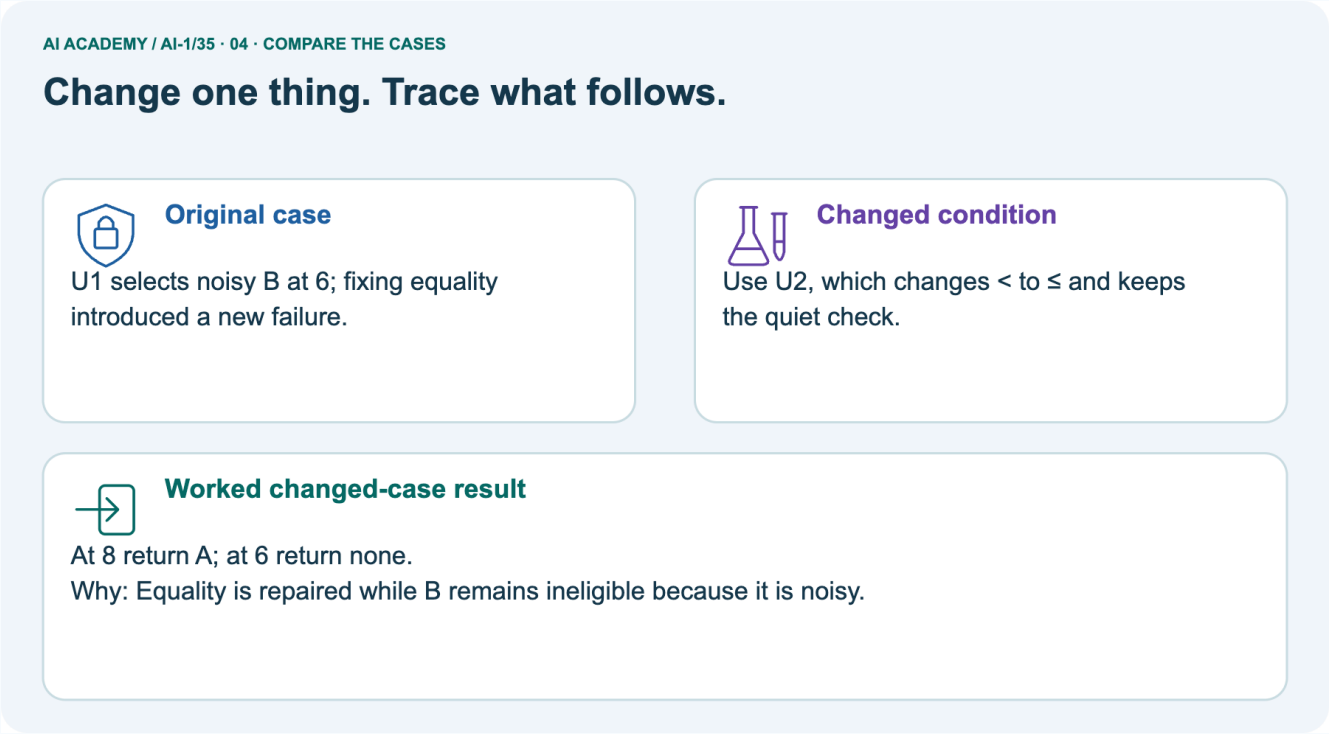
A **repair** with **evidence**. Keep the failing trace | Save a new version | Rerun the revealing case. Name the broken requirement | Correct the faulty step | Rerun earlier passing cases.

Keep **input** and catalogue labels | Record what changed | Keep every result and limit

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

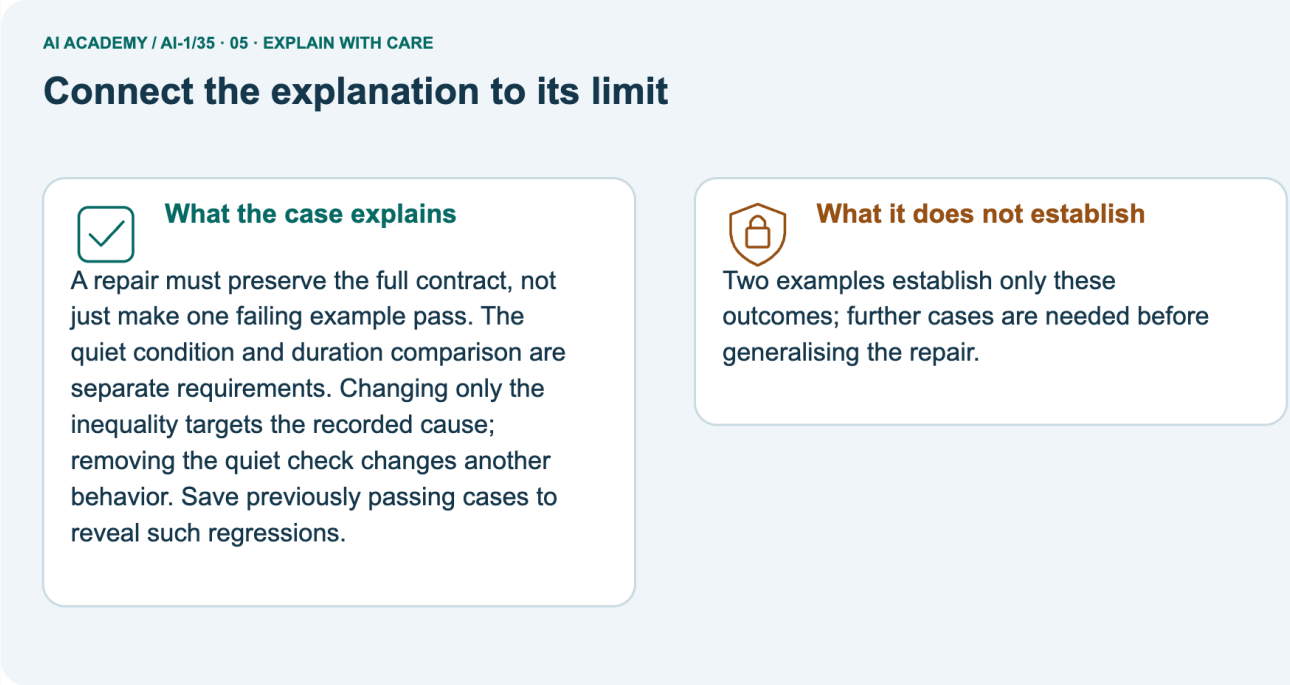
Original case: U1 selects noisy B at 6; fixing equality introduced a new failure. Changed condition: Use U2, which changes < to ≤ and keeps the quiet check. Expected result: At 8 return A; at 6 return none.

Reason: Equality is **repair**ed while B remains ineligible because it is noisy.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

A **repair** must preserve the full contract, not just make one failing example pass. The quiet condition and duration comparison are separate requirements.

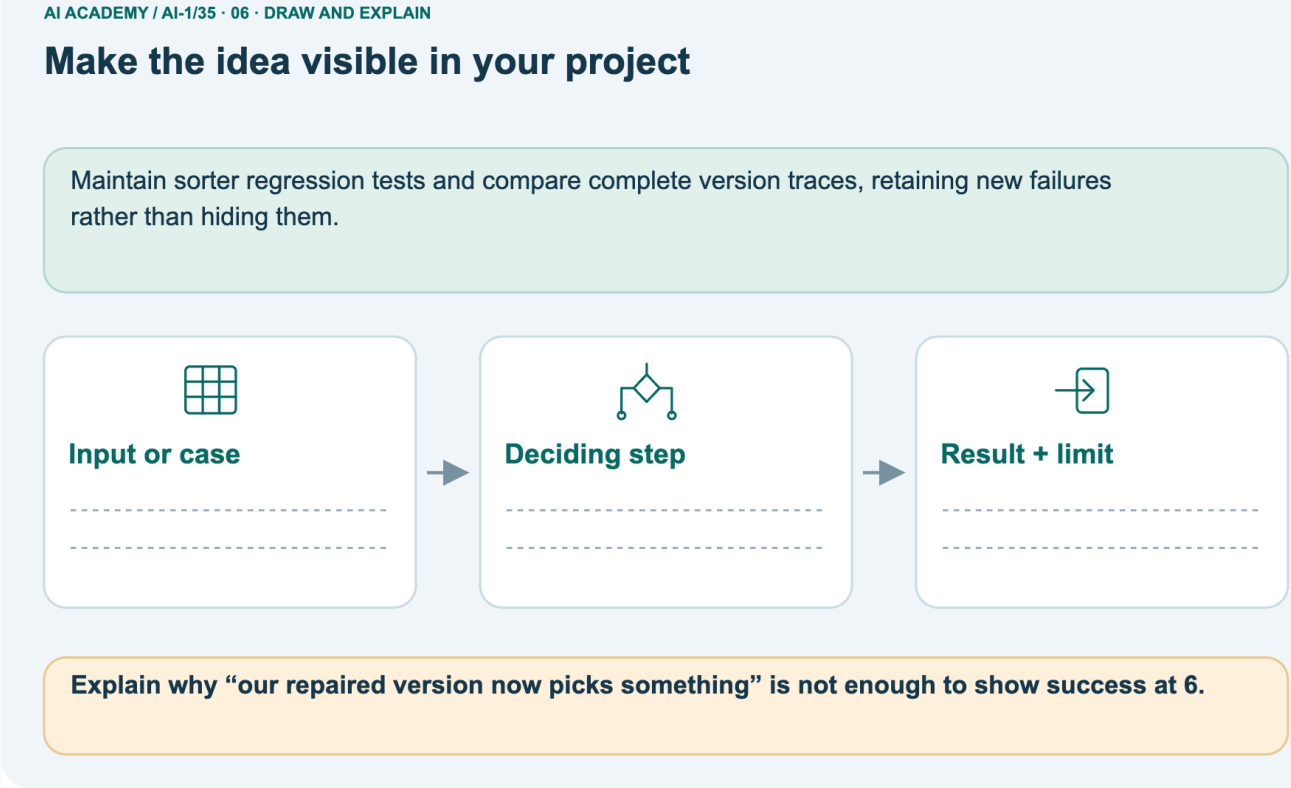
Changing only the inequality targets the recorded cause; removing the quiet check changes another behavior. Save previously passing cases to reveal such **regression**s.

Limit: Two examples establish only these outcomes; further cases are needed before generalising the **repair**.

> Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Maintain sorter **regression** tests and compare complete version traces, retaining new failures rather than hiding them. Draw your **input**, deciding step, result and limit.

Supported question: Explain why “our **repair**ed version now picks something” is not enough to show success at 6.

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.